

An Introduction to Double String Theory and Non-Riem. Backgrounds

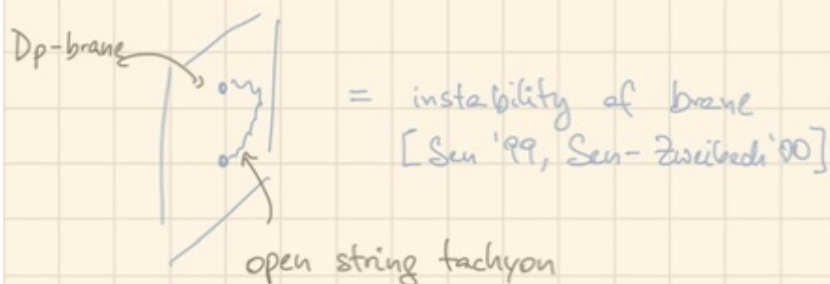
(Based on [2008.03084v2 by J.H. Park and S. Sugimoto])

PLAN

- ① Motivation: Absence of tachyon kinetic term and applications to string pheno
- ② Double Field Theory (DFT)
- ③ BRST Quantization
- ④ Case Study: non-Riem., tachyon free bosonic string

① MOTIVATION: ABSENCE OF TACHYON KINETIC TERM IN DFT

- (*) Critical bosonic string theory includes open and closed string tachyons





closed string tachyon.

? eff. description

$$S \sim \int d^D x \sqrt{-g} e^{-2\phi} \left[R + 4(\partial_\mu \phi)^2 \right.$$

$$\left. - \frac{1}{12} H_{\lambda\mu\nu} H^{\lambda\mu\nu} - \frac{2(D-26)}{3\alpha'} \right]$$

$$- (\partial_\mu T)^2 + 4/\alpha' T^2$$

(*) In non-Riem spaces it will turn out that the tachyon kinetic term vanishes

(*) In fact, we will see that the tachyon is completely absent from the closed bosonic string sector in non-Riem. space

MOTIVATION: APPLICATIONS TO STRING PHENO

(*) KK-theory was originally postulated to unify GR + EM, however one of its common drawbacks is presence of graviscalars

Consider $S \sim \frac{1}{2\kappa^2} \int_{M^{1,3} \times S^1} d^4 x d\omega \sqrt{-g} R$ with $ds^2 = \tilde{g}_{\mu\nu} dx^\mu dx^\nu + e^{2\sigma} (d\omega + A_\mu dx^\mu)^2$

$$\sim \frac{2\pi R}{2\kappa^2} \int_{M^{1,3}} d^4 x \sqrt{\tilde{g}} e^\sigma \left(R - \frac{1}{4} e^{2\sigma} F_{\mu\nu} F^{\mu\nu} + \partial_\mu \sigma \partial^\mu \sigma \right)$$

(*) If we consider compactifications on curved spaces e.g. S^2 the graviscalars get a potential as follows

$$V \propto 1/R^2$$

$\Rightarrow R \rightarrow +\infty$ minimizes the potential, so the compact space will undergo decompactification

- (o) The moduli stabilization problem persists in modern string compactifications, one approach to the problem has been to turn on fluxes or consider non-perturbative corrections

[see Kachru, Kallosh, Linde, Trivedi '03;
 Obied, Ooguri, Spodyneiko, Vafa '18;
 McAllister, Quevedo '23; ...]

- (o) However, it turns out that in compactifications on certain non-Riem. spaces the graviscalars are absent to begin with

[see Cho, Morand, Park '18]

② Double Field Theory

- (e) T-duality: a symmetry which relates physical properties corresponding to big spacetime radius with quantities corresponding to small radius
- (o) For example, consider the closed string sector mass condition with one of the physical dimensions compact. $X \sim X + 2\pi R$

$$m^2 = \underbrace{n^2/R^2}_{\text{excitation}} + \underbrace{\frac{w^2 R^2}{\alpha'^2}}_{\text{winding}} + \frac{2}{\alpha'}(N + \tilde{N} - 2)$$

- (o) In fact it turns out this duality can be written as a transformation on target space fields, which transform as an element of the infinite order discrete group $O(d, d; \mathbb{Z})$; this symmetry is manifest upon compactification
- (o) For this reason it might be useful to rewrite the action principle in variables that make the symmetry manifest, this is the goal of double field theory

$$(g_{\mu\nu}, B_{\mu\nu}, \phi) \longrightarrow (\mathcal{H}_{MN}, d)$$

$\uparrow$ $\nwarrow$
 generalised metric generalised dilaton

- (*) The indices $M, N = 0, 1, \dots, 2D$ are contracted using an $O(D, D)$ invariant metric η_{MN}

$$\eta_{MN} = \begin{pmatrix} 0_{D \times D} & 1_{D \times D} \\ 1_{D \times D} & 0_{D \times D} \end{pmatrix}$$

- (*) The defining properties are as follows

$$\mathcal{H}_{MN} = \mathcal{H}_{NM}, \quad \mathcal{H}_M{}^K \mathcal{H}_N{}^L \eta_{KL} = \mathcal{H}_{MN}, \quad e^{2d} = \sqrt{-g} e^{-2\phi}$$

The spacetime coordinates are also doubled $x^\mu \rightarrow x^M = (\tilde{x}_\mu, x^\mu)$ but physical quantities should only depend on half of these, this is achieved by a section condition

$$\partial_M \partial^M = \partial_\mu \tilde{\partial}^\mu + \tilde{\partial}^\mu \partial_\mu = 0$$

We can solve this condition via $\tilde{\partial}^\mu = 0$

- (*) A generic $O(d, d)$ rank-2 tensor has the following comp.

$$A_{MN} = \begin{pmatrix} A^{\mu\nu} & B^\mu{}_\alpha \\ B_\mu{}^\sigma & D_{\mu\nu} \end{pmatrix}$$

- (*) Finally it is not difficult to show now that in completely non-Piem. space ($\mathcal{H}^{\mu\nu} = 0$) the tachyon kinetic term that we saw previously vanishes

$$\begin{aligned} \mathcal{H}^{\mu\nu} \partial_\mu T \partial_\nu T &= \underbrace{\mathcal{H}^{\mu\nu}}_{=0} \partial_\mu T \partial_\nu T + \mathcal{H}^\mu{}_\alpha \partial_\mu T \underbrace{\tilde{\partial}^\alpha T}_{=0} \\ &\quad + \underbrace{\partial_\mu{}^\nu \tilde{\partial}^\mu T}_{=0} \partial_\nu T + \underbrace{\partial_{\mu\nu} \tilde{\partial}^\mu T}_{=0} \underbrace{\partial^\nu T}_{=0} \end{aligned}$$

- (.) Now, consider a reformulation of the bosonic string in an $O(D,D)$ manifest manner

$$S = \frac{1}{4\pi\alpha'} \int d^2\sigma \left[-\frac{1}{2} \sqrt{-h} h^{\alpha\beta} D_\alpha x^\mu D_\beta x^\nu \eta_{\mu\nu} - \epsilon^{\alpha\beta} D_\alpha x^\mu A_{\beta\mu} \right]$$

Here, D_α is a covariant derivative; x^μ is gauged by A_α^μ because it does not transform as an $O(D,D)$ vector

$$D_\alpha x^\mu := \partial_\alpha x^\mu - A_\alpha^\mu$$

We also have the constraint $A_\alpha^\mu \partial_\mu = 0$, setting $\tilde{\partial}^\mu = 0$ we can rewrite it as

$$A_\alpha^\mu \partial_\mu = A_\alpha^\mu \partial_\mu + A_{\alpha\mu} \tilde{\partial}^\mu = 0 \Leftrightarrow A_{\alpha\mu} = 0$$

- (.) A famous Ricc. parametrization of $\eta_{\mu\nu}$ is

$$\eta_{\mu\nu} = \begin{pmatrix} g^{\mu\nu} & -g^{\mu\sigma} B_{\sigma\lambda} \\ B_{\kappa\sigma} g^{\sigma\nu} & g_{\kappa\lambda} - B_{\kappa\sigma} g^{\sigma\nu} B_{\sigma\lambda} \end{pmatrix}$$

- (.) It's not difficult to show that if you insert this back to the previous Lagrangian it gives back the conventional bosonic string (set $B=0$ for simp.)

$$\begin{aligned} \mathcal{L} &= -\frac{1}{2} \sqrt{-h} h^{\alpha\beta} D_\alpha x^\mu D_\beta x^\nu \eta_{\mu\nu} - \epsilon^{\alpha\beta} D_\alpha x^\mu A_{\beta\mu} \\ &= -\frac{1}{2} \sqrt{-h} h^{\alpha\beta} (\partial_\alpha \tilde{x}_\mu - A_{\alpha\mu}) (\partial_\beta \tilde{x}_\nu - A_{\beta\nu}) \\ &\quad - \epsilon^{\alpha\beta} g^{\mu\nu} (\partial_\alpha \tilde{x}_\mu - A_{\alpha\mu}) A_{\beta\nu} - \frac{1}{2} \sqrt{-h} \partial_\alpha x^\mu \partial_\beta x^\nu g_{\mu\nu} \\ &= -\frac{1}{2} \sqrt{-h} h^{\alpha\beta} g_{\mu\nu} \partial_\alpha x^\mu \partial_\beta x^\nu \leftarrow \text{integrated out the} \\ &\quad \text{potential and fixed} \\ &\quad \text{coordinate symm. } \tilde{x}_\mu = 0 \end{aligned}$$

- (*) Ultimately we will be interested in non-Piem. spaces in these spaces the generalised metric $\mathcal{G}_{\mu\nu}$ can be parametrised in terms of

$$(X_\mu^i, \bar{X}_\mu^i) \text{ and } (Y_i^\nu, \bar{Y}_i^\nu)$$

$$\text{satisfying } X_\mu^i Y_i^\nu + \bar{X}_\mu^{\bar{i}} \bar{Y}_{\bar{i}}^\nu = \delta_\mu^\nu$$

Here, $i = 1, 2, \dots, n$ and $\bar{i} = 1, 2, \dots, \bar{n}$; $n + \bar{n} = D$.

Assuming $B = 0$ for simplicity we find

$$\mathcal{G}_{\mu\nu} = \begin{pmatrix} 0 & Y_i^\mu X_\nu^i - \bar{Y}_{\bar{i}}^\mu \bar{X}_{\bar{i}}^\nu \\ X_\mu^i Y_i^\nu - \bar{X}_{\bar{i}}^\mu \bar{Y}_{\bar{i}}^\nu & 0 \end{pmatrix}$$

- (*) We can insert this back to the action to find

$$\begin{aligned} \mathcal{L} &= -\frac{1}{2} \sqrt{-h} h^{\alpha\beta} D_\alpha x^\mu D_\beta x^\nu \mathcal{G}_{\mu\nu} - \mathcal{E}^{\alpha\beta} D_\alpha x^\mu A_{\beta\mu} \\ &= -\frac{1}{2} \sqrt{-h} h^{\alpha\beta} (\partial_\alpha \tilde{x}_i - A_{\alpha i}) \partial_\beta x^i + \frac{1}{2} \sqrt{-h} h^{\alpha\beta} (\partial_\alpha \tilde{x}_{\bar{i}} - A_{\alpha \bar{i}}) \partial_\beta x^{\bar{i}} \\ &\quad - \frac{1}{2} \sqrt{-h} h^{\alpha\beta} \partial_\beta x^i (\partial_\alpha \tilde{x}_i - A_{\alpha i}) + \frac{1}{2} \sqrt{-h} \dots \\ &= -2\sqrt{-h} (P_{\alpha i} h^{\alpha\beta} \partial_\beta x^i + P_{\alpha \bar{i}} h^{\alpha\beta} \partial_\beta x^{\bar{i}}) + \mathcal{E}^{\alpha\beta} \partial_\alpha \tilde{x}_i \partial_\beta x^i \\ &\quad \uparrow \\ &\quad P_{\alpha i} = \partial_\alpha \tilde{x}_i - A_{\alpha i} \quad \text{end} \quad h^{\alpha\beta}_{\pm} = \frac{1}{2} (h^{\alpha\beta} \pm \frac{\mathcal{E}^{\alpha\beta}}{\sqrt{-h}}) \end{aligned}$$

③ BRST QUANTIZATION [Following Polchinski I]

- (*) BRST quantization allows us for a systematic quantization, especially if we want to keep track of potential anomalies

- (.) Consider a path integral with fields ϕ_i and gauge symm. $\varepsilon^\alpha \mathcal{S}_\alpha$, here by assumption the gauge transformations have to satisfy an algebra

$$[\mathcal{S}_\alpha, \mathcal{S}_\beta] = f_{\alpha\beta}^\gamma \mathcal{S}_\gamma$$

Typically, next we would like to enforce a certain gauge $F(\phi) = 0$

- (.) As always, we can enforce this gauge dynamically via the Faddeev-Popov procedure

$$\int \frac{[d\phi]}{V_{\text{gauge}}} \exp(-S_1) \rightarrow \int [d\phi] [db_A] [dc^\alpha] \exp(-S_1 - S_2 - S_3)$$

$$\text{with } S_2 = -i B_A F^A(\phi), S_3 = b_A c^\alpha \mathcal{S}_\alpha F^A(\phi)$$

- (.) The set above admits what's known as BAST symmetry

$$\mathcal{S}_B \phi_i = -i \varepsilon c^\alpha \mathcal{S}_\alpha \phi_i,$$

$$\mathcal{S}_B B_A = 0,$$

$$\mathcal{S}_B b_A = \varepsilon B_A,$$

$$\mathcal{S}_B c^\alpha = \frac{i}{2} \varepsilon f_{\beta\gamma}^\alpha c^\beta c^\gamma$$

Since $\mathcal{S}_B$ acts akin to a gauge trans. on $\phi_i \Rightarrow \mathcal{S}_B S_1 = 0$

- (.) Now, let's consider the effect of a small change in our gauge fixing condition δF on a correlation func.

$$\begin{aligned} \varepsilon \delta \langle f | i \rangle &= \langle f | -\varepsilon \mathcal{S} (S_2 + S_3) | i \rangle \\ &= i \langle f | \mathcal{S}_B (b_A \mathcal{S} F^A) | i \rangle \end{aligned}$$

$$= -\epsilon \langle f | \{ Q_B, b_A \delta F^A \} | i \rangle$$

For the above to hold for arbitrary δF we must have

$$Q_B | \psi \rangle = Q_B | \psi' \rangle = 0$$

(.) The take-away is that physical states must be BRST invariant

(o) In order to move around in the space of gauge choices, the BRST operator must remain conserved; thus, it must commute with the change in the Hamiltonian

$$\begin{aligned} 0 &= [Q_B, \{ Q_B, b_A \delta F^A \}] \\ &= Q_B^2 b_A \delta F^A - Q_B b_A \delta F^A Q_B + Q_B b_A \delta F^A Q_B \\ &\quad - b_A \delta F^A Q_B^2 \\ &= [Q_B^2, b_A \delta F^A] \end{aligned}$$

(.) For general gauges this implies $Q_B^2 = 0$

(.) To conclude the discussion of BRST Quantization let me mention the concept of BRST cohomology

We call a state $Q_B | \chi \rangle$ physical since it is annihilated by $\hat{\omega}_B$

$$Q_B (Q_B | \chi \rangle) \rightarrow Q_B (Q_B | \chi \rangle)$$

physical + null

(.) However, it is also a null state since its dot product with any other physical state is zero, including itself

So any two states that differ by a null state are physically equivalent

(0) The physical spectrum lies in the BRST cohomology

$$\mathcal{H}_{\text{BRST}} = \frac{\mathcal{H}_{\text{closed}}}{\mathcal{H}_{\text{exact}}}$$

states annihilated by Q_B

states analogous to $Q_B |X\rangle$

④ CASE STUDY: NON-RIEM., TACHYON FREE BOSONIC STRING

(0) The action in the non-Riem. space was

$$\mathcal{L}_0 = -\sqrt{-h} [P_{\alpha i} h^{\alpha\beta}_+ \partial_\beta x^i + P_{\alpha \bar{i}} h^{\alpha\beta}_- \partial_\beta x^{\bar{i}}] + \epsilon^{\alpha\beta} \partial_\alpha \tilde{x}_\mu \partial_\beta x^\mu$$

(0) Let's change variables to

$$\sqrt{-h} h^{\tau\tau} = -1/e, \quad \sqrt{-h} h^{\tau\sigma} = \omega/e, \quad \sqrt{-h} h^{\sigma\sigma} = e - \omega^2/e$$

Let's also move to light cone coordinates

$$\sigma^\pm = \tau \pm \sigma, \quad \partial_\pm = 1/2 (\partial_\tau \pm \partial_\sigma),$$

$$c^\pm = c^\tau \pm c^\sigma, \quad P_{\pm\mu} = 1/2 (P_{\tau\mu} \pm P_{\sigma\mu})$$

(.) The action we wrote is completely covariant under symmetries including:

O(0, D) notations, Weyl symmetry,

worldsheet diffs, doubled target space diffs

- (0) We can proceed to fix coordinate gauge symmetry and worldsheet diffs

In fact the previous Lagrangian also admits an additional symmetry

$$\delta p_{\alpha i} = h_{+\alpha}{}^B \hat{C}_{Bi}, \quad \delta p_{\alpha \tau} = h_{-\alpha}{}^B \hat{C}_{B\tau}$$

$\uparrow$
arbitrary local parameters

- (0) We can fix all the gauges via

$$e = 1, \quad \omega = 0, \quad \tilde{x}_\mu = 0, \quad p_i = 0, \quad p_{+\tau} = 0$$

The full Lagrangian with the Faddeev-Popov ghosts is

$$\mathcal{L}_{\text{full}} = \mathcal{L}_0 - i \delta_B (b_e \ln e + \omega b_\omega + \tilde{x}_\mu \tilde{B}^\mu + p_{-i} B^i + p_{+\tau} B^\tau)$$

simplify

$$\mathcal{L}_{\text{red}} = 2 (p_{+i} \partial_- x^i + p_{-\tau} \partial_+ x^\tau + i b_{++} \partial_- c^+ + i b_{--} \partial_+ \bar{c})$$

Setting $b_e = b_{++} + b_{--}$, and $b_\omega = b_{++} - b_{--}$

- (0) We can identify $\{p_{+i}, p_{-\tau}\}$ as conjugate momenta of $\{x^i, x^\tau\}$ forming D-pairs of a 'BD' system with weights 1 and 0

$$Q_B = Q_L + Q_R$$

$$Q_L = \oint ds (P: \partial_+ x^i) c^+ + i (b_{++} \partial_+ c^+) c^+$$

$$= : \sum_{m,n=-\infty}^{\infty} m (-i \beta_{mi} x_n^i + b_m c_n) \epsilon_{-m-n} : - a c_0$$

The quantization proceeds as usual,

with $[\partial_m^i, p_n^j] = i \delta_{ij} \delta_{m+n}$ and $\{b_m, c_n\} = \delta_{m+n}$

- (.) It turns out that the BRST charges are Q_L, Q_R nilpotent only when $n = \bar{n}$

This implies that the usual critical dimension

$$D = 26 \text{ since } c_L = 2n - 26, c_R = 2\bar{n} - 26$$

- (.) The physical states are annihilated by Q_L and the antighost zero mode

$$L_0 = \{b_0, Q_R\} = N_B + N_X + N_b + N_c - a$$

Why by b_0 ?

the origin is kinetic

see p. 131 Polchinski 1

All the level counting operators are positive semi-definite, implying a drastic simplification of the spectrum to just one level

- (e) Since $\langle 0 | [L_{-1}, L_1] | 0 \rangle = -2$ with $L_n = \{Q_L, b_n\}$,

we can identify the level to be unity

Thank You !